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PALAEOMAGNETIC AND RADIOMETRIC EVIDENCE FOR THE AGE OF THE FREETOWN IGNEOUS COMPLEX, SIERRA LEONE

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New K/Ar whole rock age determinations at 3 localities and palaeomagnetic studies at 11 localities in the Freetown layered igneous complex, are reported and discussed in the light of previous geological and radiometric evidence for the age of the complex. Analysis of NRM reveals two components – one very large but magnetically soft is believed to be IRM, the other rather weaker but extremely stable is believed to be thermoremanent because it has very similar properties to TRM imposed in the laboratory. This stable magnetization has reversed polarity (D = 175°, I = −11°) at the three localities nearest to the northeastern margin, and hence nearest the base of the layered complex. Elsewhere normal polarity is observed (D = 350°, I = + 2°). NRM/TRM ratios are lowest close to the boundary between areas of normal and reversed polarity. Geological evidence reinforces the view that this stable NRM is primary and that the dual polarity is the result of geomagnetic field reversal. The axis of stable magnetization is D = 352°, I = + 7°, k = 43, α_{95} = 7° as deduced from a.f. cleaning of specimens from N = 10 localities, (one locality was discarded because of anomalous magnetization). The corresponding palaeomagnetic pole is lat. 81° N, long 135° WW, A_{95} = 4.8° which is only matched in Africa by poles of Triassic to Lower Jurassic age.

Previous radiometric ages from the complex ranged from 190 to 1875 my, and determinations of 185 my and 230 my had been obtained from dykes of a swarm which cuts the complex. New K/Ar whole rock analyses reported here yield a mean age of 180 my which may be taken as a minimum age for the complex. Taken together, the palaeomagnetic and radiometric evidence indicate that the complex was probably emplaced during the Triassic period.

1. Introduction

Until recently, all African pre-Tertiary palaeomagnetic field directions have been derived from south and east Africa (a study in the Precambrian of Mauretania by Gross and Strangway [1] having produced no results in terms of the ancient geomagnetic field). As a first step towards extending the coverage to west Africa, one of us (D.I.H.) collected eleven samples of gabbroic rocks from the Freetown Igneous Complex, Sierra Leone to test the feasibility of palaeomagnetic study and to investigate the anomalous radiometric age determinations from the complex [2, 3] which ranged between 190 and 1875 my. A successful palaeomagnetic and geochronological study would enable comparison to be made with south African data in a test of the geocentric dipole field model if the age were shown to be Phanerozoic, or, in the event of a Precambrian age being determined, an assessment of the relative positions of western and southern Africa.

The Freetown Complex is a layered basic igneous intrusion outcropping to form the peninsula south of Freetown. Both gravity survey [4] and the layered structure of the complex [5] indicate a roughly circular outline centred some 15 km offshore. The gravity profile, and boreholes penetrating the complex into the Precambrian basement near its eastern margin both indicate that the intrusion is funnel shaped or lopolithic.

2. Sampling and measurement procedure

Because the natural surface is deeply weathered, sampling was confined to stream beds, road cuts and

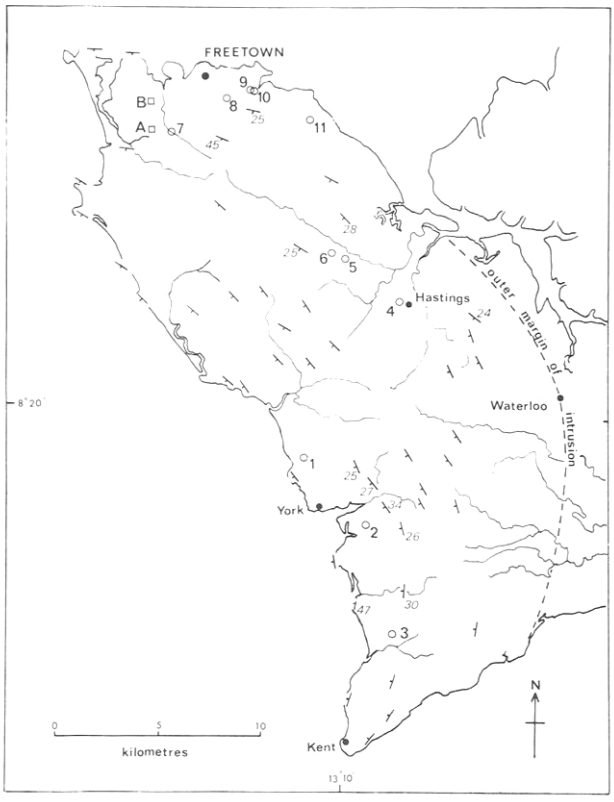


Fig. 1. Geological sketch map of the Freetown igneous complex from Wells [5] showing palaeomagnetic sampling localities (○) and the location of radiometric samples (□). The plane of igneous lamination is shown, but its dip has only been marked in the vicinity of the palaeomagnetic sites.

a quarry; their localities are shown in fig. 1. One hand sample was collected at each locality and oriented with clinometer and sun compass wherever possible (magnetic bearings had to be substituted at sites [8–11]). Direction and intensity of magnetization were measured with a P.A.R. SM2–D spinner magnetometer, and a.f. demagnetization employed equipment of the type described by McElhinny [6]. TRMs were imposed by heating in a muffle furnace, and cooling the specimens in the geomagnetic field in the laboratory.

3. Palaeomagnetic directions and their analysis

3.1. Total NRM

Total NRM directions (fig. 2a) are consistent within sites, but agreement between sites is rather poor ($k = 2.5$) and the calculated mean direction ($N = 11$ sites) is $D = 335°$, $I = -7°$, $\alpha_{95} = 37°$ (table 2). Initial intensities ranged from 0.1 to 32 × 10^{-3} gauss.

3.2. Alternating field demagnetization and magnetic cleaning

During progressive a.f. demagnetization, rapid changes in both intensity and direction of NRM (figs. 2b, 3a) take place with applied fields of only a few hundred oersteds.

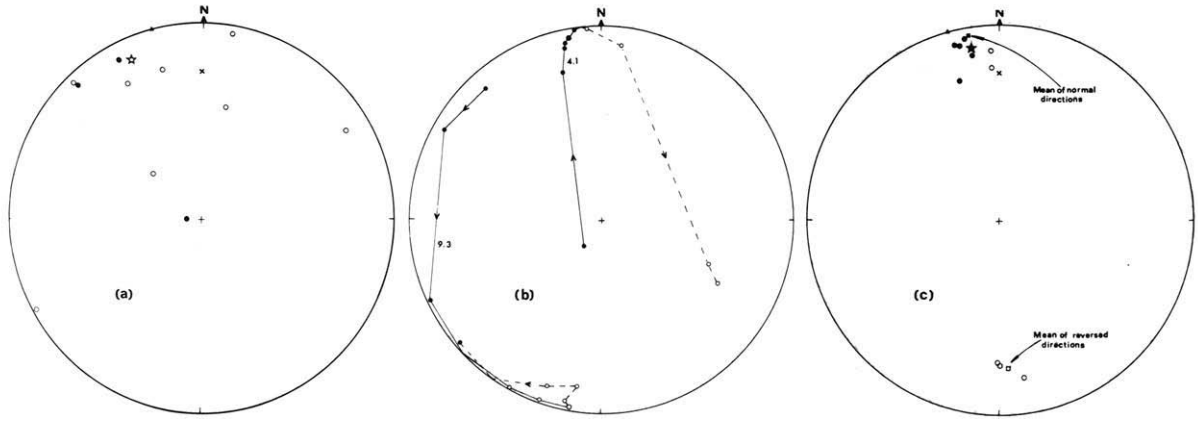


Fig. 2. Stereographic projections of (a) site mean directions of total NRM; (b) changes of NRM direction during progressive demagnetization of specimens from localities 4 and 9; (c) site mean directions after a.f. cleaning. Open symbols denote upward (-ve) inclination and solid symbols denote downward (+ve) inclination. In (a) and (c) the overall mean is denoted by a star, the present geomagnetic field by a triangle and the axial dipole field by a cross. In (b) the direction after magnetic cleaning in the optimum alternating field, is denoted by a slightly larger symbol.

Table 1
Palaeomagnetic site statistics.

Site	Total NRM				Optimum field for magnetic cleaning (oe peak)	After magnetic cleaning			
	No. of specimens	R	D	I		No. of specimens	R	D	I
1	4	4.00	310	−51	500	2	2.00	357	−8
2	3	2.99	316	−2	300	2	2.00	357	−14
3	4	3.99	9	−3	500	2	2.00	345	+18
4	4	3.95	264	+80	500	2	2.00	349	+4
5	3	3.00	316	+4	1000	2	2.00	347	+5
6	3	2.97	58	−7	500	2	2.00	346	+5
7	4	3.17	344	−14	–				
8	3	2.98	332	+5	500	2	2.00	352	+9
9	3	2.60	330	−15	800	2	1.90	180	−17
10	4	3.37	241	−2	500	2	1.97	171	−12
11	3	3.00	12	−30	1000	2	1.96	180	−19

Table 2
Summary of palaeomagnetic directions.

		No. of sites (N)	R	D	I	k	α_{95}
Total NRM		11	7.04	335.1	−7.1	2.5	36.6
a.f. cleaned	normal	7	6.87	350.4	2.5	47	8.9
	reversed	3	2.97	176.6	−15.9	69	15.0
	combined	10	9.79	352.3	6.5	43	7.4

At higher fields, the curve flattens and changes of −direction are small. These stable directions are grouped around $D = 350°, I = +2°$ (sites 1–6 and 8) or $D = 177, I = -16$ (sites 9–11) (fig. 2c, tables 1 and 2). Coercivities of this stable component of NRM range from a few hundred to 1600 Oe or more – in many cases, up to the maximum field generated by the demagnetizer. In other cases, a further decrease in intensity is observed at these very high fields and is accompanied by a scattering of directions.

At site 7, the a.f. demagnetization curves for the two specimens are completely different, and the grouping of directions after every stage is random; no stable end direction is reached, and an unusually high proportion of NRM has coercivity greater than 2000 Oe. The origin of NRM at site 7 is therefore uncertain, and this site is excluded from further consideration.

Briden [7] has defined a stability index which reaches its maximum in the remanence coercivity range for which successive changes in the NRM vector are minimum. The maximum stability index thus coincides with optimum separation of unstable and stable components. The optimum fields for magnetic cleaning at each site chosen by this method vary between sites from 300 to 1000 Oe (peak) (fig. 3).

After magnetic cleaning, the north-seeking directions at the seven normally magnetized sites and the south seeking directions at the three reversed sites were combined in calculating an overall mean axis of magnetization for the intrusion ($D = 352°, I = +7°$, $\alpha_{95} = 7°$). This procedure is justified because Watson's test [8] shows that the reversal is not significantly different from 180°.

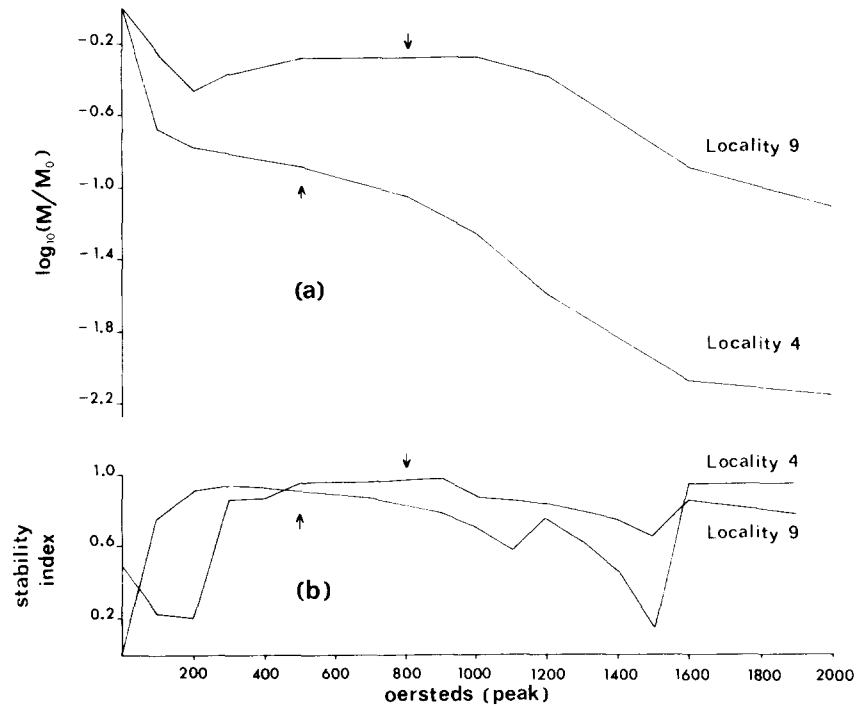


Fig. 3. (a) A.f. demagnetization curves. (b) Corresponding stability index curves [7]. The optimum field for magnetic cleaning is indicated by the arrows.

3.3. Relation of magnetism to igneous lamination

The presence of igneous lamination throughout the complex offers the unusual opportunity to carry out a palaeomagnetic fold test in intrusive rocks, though in this case the fold test criteria are inverted in comparison with the usual sedimentary situtation. The precision of grouping of stable NRM directions relative to their present in situ positions is estimated as $k = 43.4$. If the lamination is restored to the horizontal by rotation about its present strike at each sampling location, then $k = 23.7$. This difference is marginally significant at the 90% probability level using F-ratio tables to assess significance [9]. This test indicates that the stable NRM was probably acquired when the igneous lamination was in the same configuration as it is now; i.e. either NRM and dip of lamination are both primary or they are both secondary. It is argued strongly here and in sect. 4 that the NRM is of primary origin, and consequently we regard the dip of the igneous lamination to be predominantly a crystallisation phenomenon.

4. NRM/TRM ratios and the origin of NRM

There are two main lines of evidence for the origin of the NRM and the occurrence of normal and reversed polarity: namely the magnetic mineralogy and comparison of NRM with a laboratory imposed TRM.

Petrological studies by Wells [5] suggest three possible sources of remanent magnetization: magnetite grains containing exsolved ilmenite lamellae, ilmenite grains containing exsolved magnetite lamellae, and elongate magnetite grains exsolved along cleavage planes in clinopyroxenes. Since equidimensional magnetite grains have coercivities of only a few hundred oersteds [10], the high coercivity component must originate in exsolved magnetite (either within ilmenite or pyroxenes) rather than in the magnetite crystals themselves. Elongate crystals of magnetite, by contrast, can have coercivities up to several thousand oersteds [11] and will behave as single domain grains provided their axial ratios are in excess of 1:10.

To investigate the possible modes of magnetiza-

tion, the a.f. coercivity spectrum of NRM was compared with that of a TRM imposed by heating the specimen to 700°F and cooling in the earth's fields in the laboratory (0.42 Oe). For comparison, the demagnetization curves (fig. 4) are normalised to the optimum field for a.f. cleaning as determined in sect. 3. Good agreement is observed for fields greater then the optimum, which suggests that the high coercivity component of NRM is of thermoremanent origin. Below the optimum field the curves diverge which implies a different origin for the low coercivity NRM. High intensity/low coercivity remanence is usually IRM.

From the thermoremanent component of NRM, estimates of the palaeointensity (F_p) of the geomagnetic field at Freetown may be made on the assumption that $F_p = M_{NRM}/M_{TRM} \cdot F_t$ (with the notation of fig. 5).

Profiles of NRM/TRM versus distance from the estimated edge of the intrusive (fig. 5) show a minimum near the inferred boundary between areas of reversed and normal polarity. This minimum may therefore correspond to a minimum in ancient geomagnetic field intensity during a polarity transition.

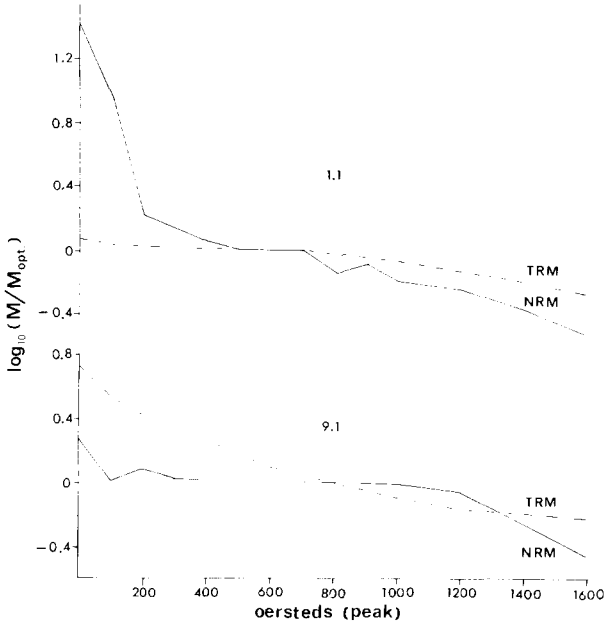


Fig. 4. A.f. demagnetization curves of NRM and TRM for two specimens, normalised to their respective optimum a.f. cleaning fields.

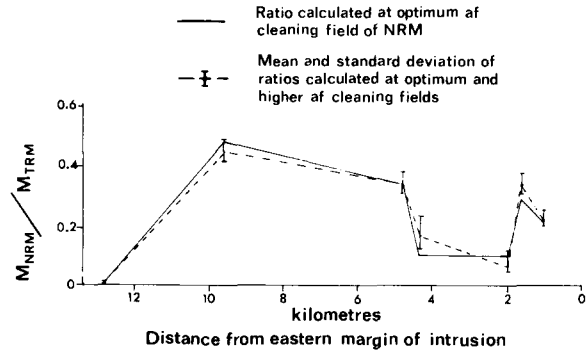


Fig. 5. NRM/TRM ratio versus estimated distance from eastern edge of the intrusion. The TRM was imposed in the geomagnetic field (F_t) in the laboratory. M_{NRM}, M_{TRM} are the remanent magnetizations after a.f. cleaning in equal fields as indicated in the figure.

In the exposed segment of the complex, distance from the margin is believed to be roughly proportional to "stratigraphic" height above its base. This polarity transition, preserved within 1 km of the inferred base, has thus been recorded during upward migration of the Curie isotherm.

Another low NRM/TRM ratio is plotted on the left hand edge of fig. 5. It is derived from only one sample, from a locality which is likely to be much more than 1 km from either top or bottom of the complex. We are inclined to think that this low ratio is related to a second polarity transition, rather than resulting from a "mirror image" record of the same transition due to cooling from the roof of the complex.

5. The age of the complex

To the east, the complex is overlain by Cenozoic sediments of the Bullom series, the oldest palaeontological evidence from which indicates a Lower Eocene age. The intrusion itself is cut by many WNW trending dykes, part of a swarm which is widespread in Sierra Leone, and which elsewhere has yielded ages of 185 and 230 my [2]. K/Ar whole rock age determinations (table 3) on three samples from the complex yield ages ranging from 165 my to 194 my which could approximate to the age of intrusion or

Table 3
Potassium-argon whole rock age determinations, Freetown Complex.

Sample	Locality (fig. 1)	K %	Vol ^{40}Ar rad. scm^3/g × 10^{-5}	^{40}Ar rad. %	Age (my)
8524	A	0.19	0.1306	63.2	165 ± 10
8527	B	0.33	0.2374	62.2	172 ± 8
			0.2301	63.1	
8631	B	0.29			194 ± 8
			0.2440	75.8	

$\lambda_\beta = 0.472 \times 10^{-9}$ yr^{-1}; $\lambda_e = 0.584 \times 10^{-9}$ yr^{-1}; ^{40}K/K = 0.0119 atomic %.

may, like the determinations on the dykes, merely define a minimum age for the intrusion complex.

A maximum age of 300 ± 20 my for the intrusion is suggested by a K/Ar determination on hornblende from borehole S.L.B.H. 9 at depth 391 to 394 ft in the Kasila Gneiss immediately below the intrusion [3]. This apparent age is probably the result of partial expulsion of argon from the Precambrian gneiss due to thermal effect of intrusion. Moreover, there is no evidence of the Pan-African thermal and orogenic event having affected the intrusion, although it has affected the basement.

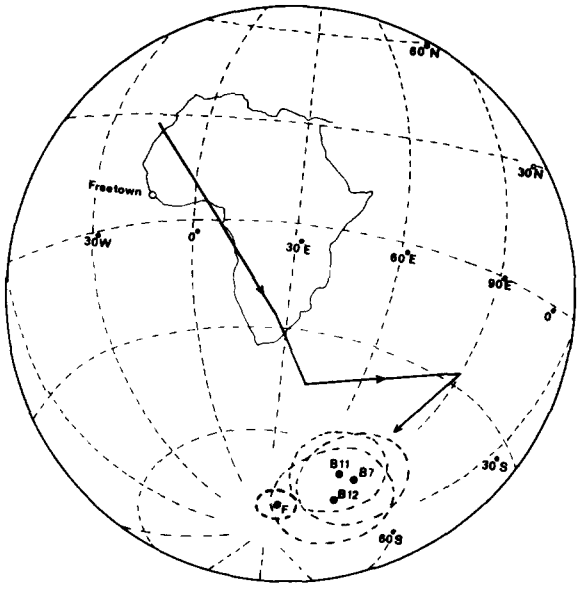


Fig. 6. Palaeomagnetic pole (F) from the Freetown Complex, compared with the Palaeozoic and Mesozoic polar wander curve for Africa [12] and with the poles having 95 per cent errors which overlap or nearly touch that of Freetown: B7: Shawa ijolite, Rhodesia [13], B12: Stormberg lavas [14], B11: Karroo dolerites [15].

Radiometric age determinations from pyroxenes and plagioclases from the igneous complex range from 180 my to 1875 my [2]. Snelling [2] suggests that the oldest dates are more realistic and that younger dates indicate argon loss. Andrews-Jones [3], however, argues for a Mesozoic age for the intrusion and associates the older ages with excess argon acquired from the country rocks during emplacement. This view is favoured by the present authors and is supported by the palaeomagnetic evidence. It is also consistent with the conclusions of Wells (ref. [5], pp. 94–95) that the dolerite dykes (which are now known to be Mesozoic) formed the last stage in the same series of magmatic events as the emplacement of the layered complex itself.

The pole position determined from this survey 80.9°S, 44.6°E, with $A_{95} = 4.8°$ as calculated from the $N = 10$ samples after magnetic cleaning (table 2), is matched by no known Precambrian pole from Africa. Its circle of 95 per cent confidence intersects that of only one Phanerozoic pole from Southern Africa regarded as reliable by McElhinny et al. [12] (fig. 6) and almost touches the circles of confidence of two others. All three of these are Triassic or early Jurassic (say 220 to 160 my).

Thus radiometric evidence provides an approximate estimate of 180 my for the *minimum* age of the Freetown complex, while palaeomagnetism effectively indicates an *older* limit of about 220 my. It is therefore concluded that the complex was probably emplaced during the Triassic period.

Acknowledgements

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